

Status of Benzene Emissions and Monitoring in the UK

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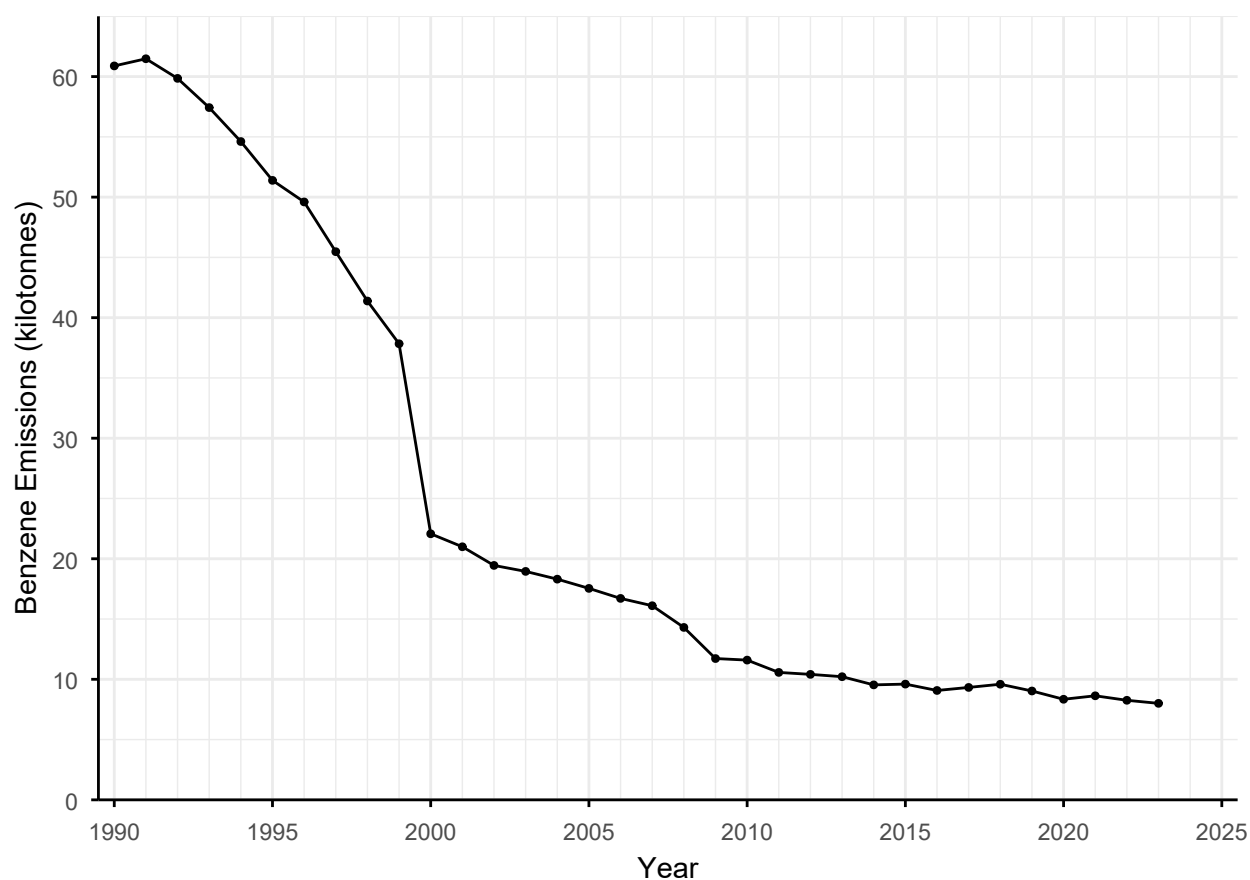
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Total Emissions

The National Atmospheric Emissions Inventory (NAEI) produce annual estimates on the emission of Benzene in the UK. Total emissions have fallen from 60.9 kt in 1990 to 8.0 kt in 2023, a reduction of 87%. The majority of this reduction happened in the 1990s, as catalytic converters and evaporative emission controls spread through the petrol car fleet. The single biggest year-on-year drop was between 1999 and 2000, when emissions fell from 37.8 kt to 22.1 kt. This coincided with the EU Fuel Quality Directive (98/70/EC) reducing the maximum allowable benzene content of petrol from 5% to 1% by volume from January 2000. Since around 2010 annual emissions of benzene have not maintained a steep reduction trend.

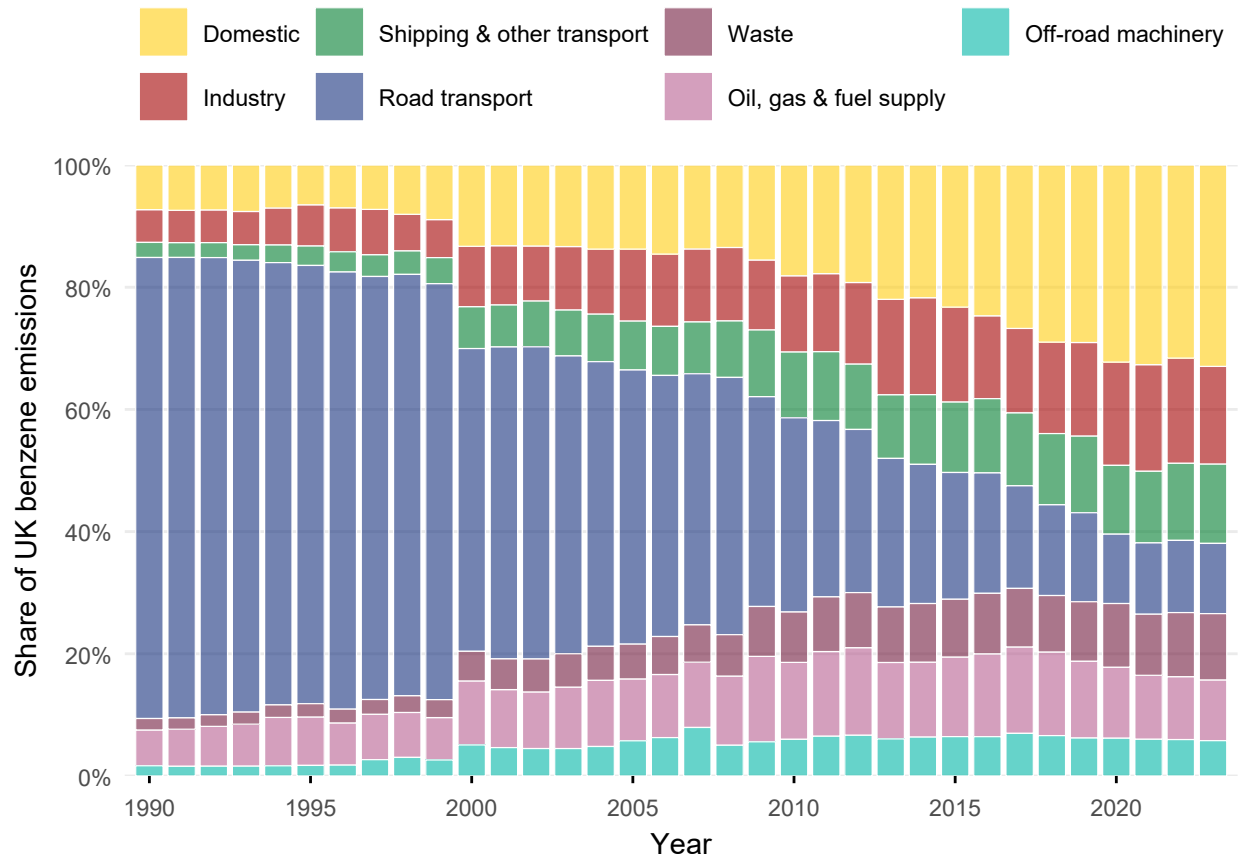
Emissions reported as memo items (including international maritime navigation and international aviation cruise, “z_”) were excluded, consistent with national totals reported under the UNECE Convention on Long-range Transboundary Air Pollution (CLRTAP).



By Emission Source

The NAEI reports benzene emissions from about 342 individual sources. For this report they have been grouped into seven broad categories (see `benzene_source_lookup.csv` for the full mapping). The figure below shows each category's share of the national total in each year.

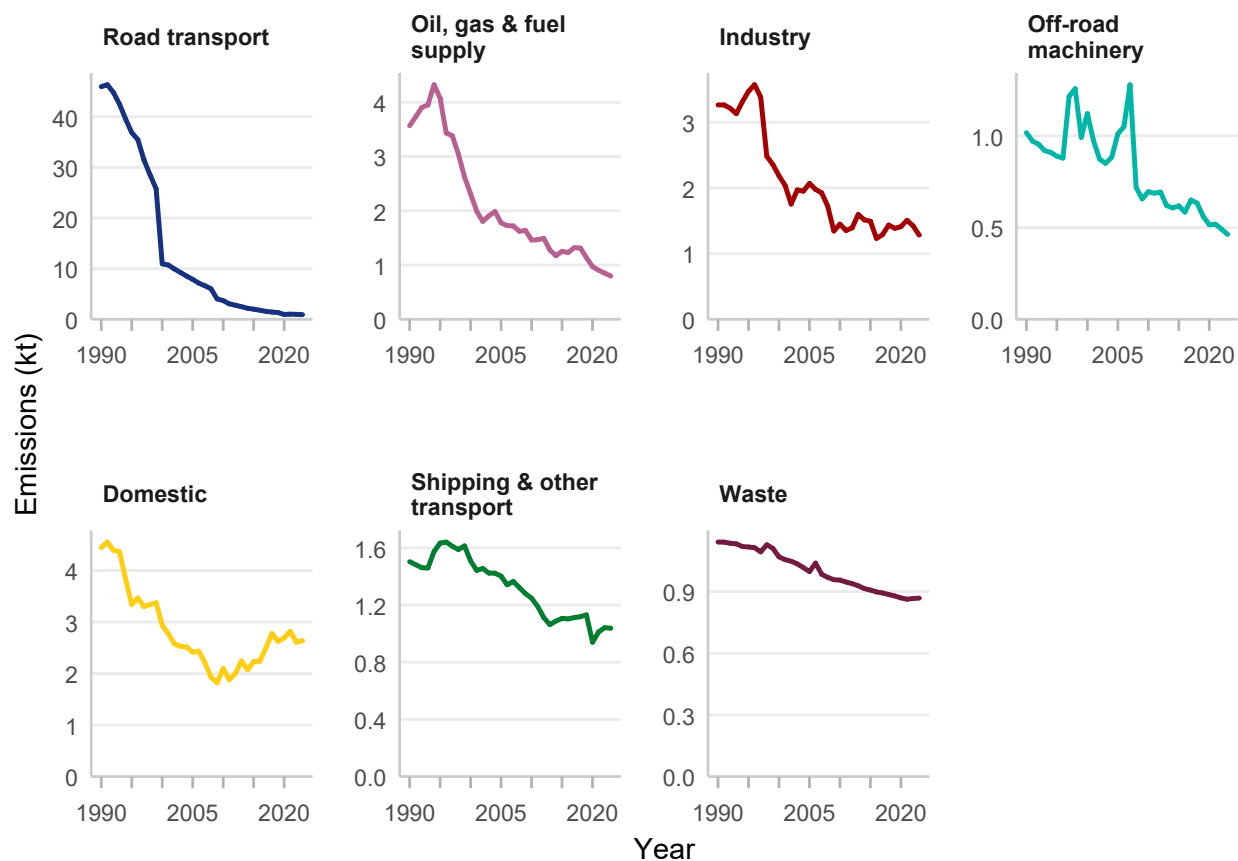
In 1990, road transport dominated UK benzene emissions and accounted for three-quarters (75%) of the total. Its share has fallen steadily as regulation has been introduced, and by 2023 road transport contributed 12% of the total emissions (0.9 kt, down from 46 kt). No other source have seen similar levels of reduction so the mix of sources now looks different. In 2015, Domestic combustion overtook road transport as the largest single source in 2015 and made up a third (33%) of emissions in 2023. Industry (16%), shipping and other transport (13%) and waste (11%) are now all comparable to or larger than road transport.



However, a rising share does not mean rising emissions. Domestic emissions fell from 4.4 kt in 1990 to 2.6 kt in 2023; they simply fell more slowly than everything else. The next figures show the change in absolute terms. Within the domestic sector, the fuel mix has changed markedly. Coal burning, which produced 1.9 kt in 1990, now produces less than 0.1 kt. Over the same period, emissions from domestic wood burning have roughly tripled, from 0.5 kt to 1.7 kt, and wood is now the main source of domestic benzene. Waste and shipping have also become relatively more important as their emissions have changed little since 1990.

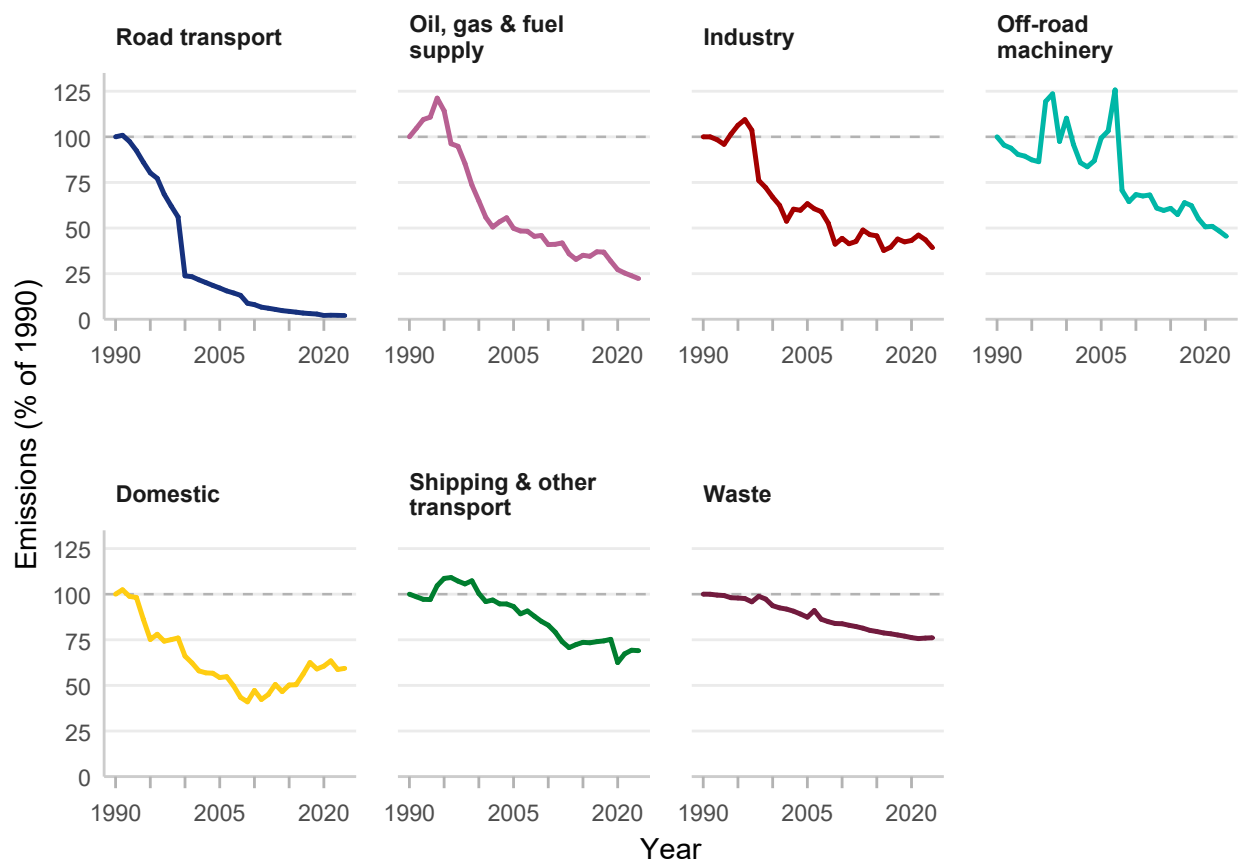
Absolute

Each panel shows benzene emissions from one source category in kilotonnes. *Note:* Every panel has a separately scaled y-axis.



Relative to baseline

Each line shows a category's emissions as a percentage of the 1990 emissions. Road transport has fallen to 2% of its 1990 level, while waste (76%), shipping and other transport (69%) and domestic combustion (59%) have reduced by a lower fraction. These sources now make up a much bigger share of the total.

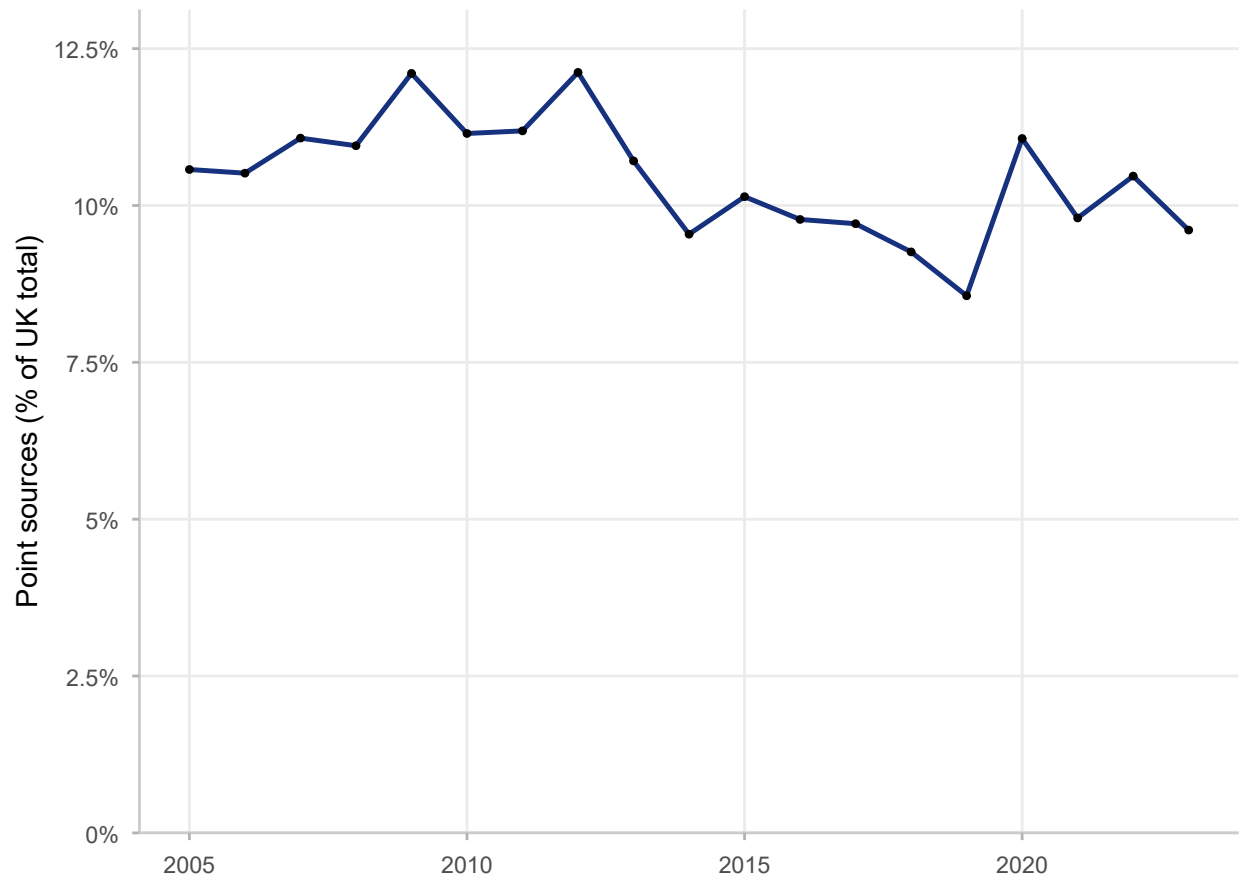


These figures show that UK benzene emissions have changed from being dominated by a mobile distributed source to being spread across several smaller ones. Many of the remaining sources, such as refineries and industrial plants, are located in fixed places rather than spread along the road network. This means national totals may say less about local exposure than they once did, and it motivates looking at where emissions occur and what we are monitoring - discussed in the next section.

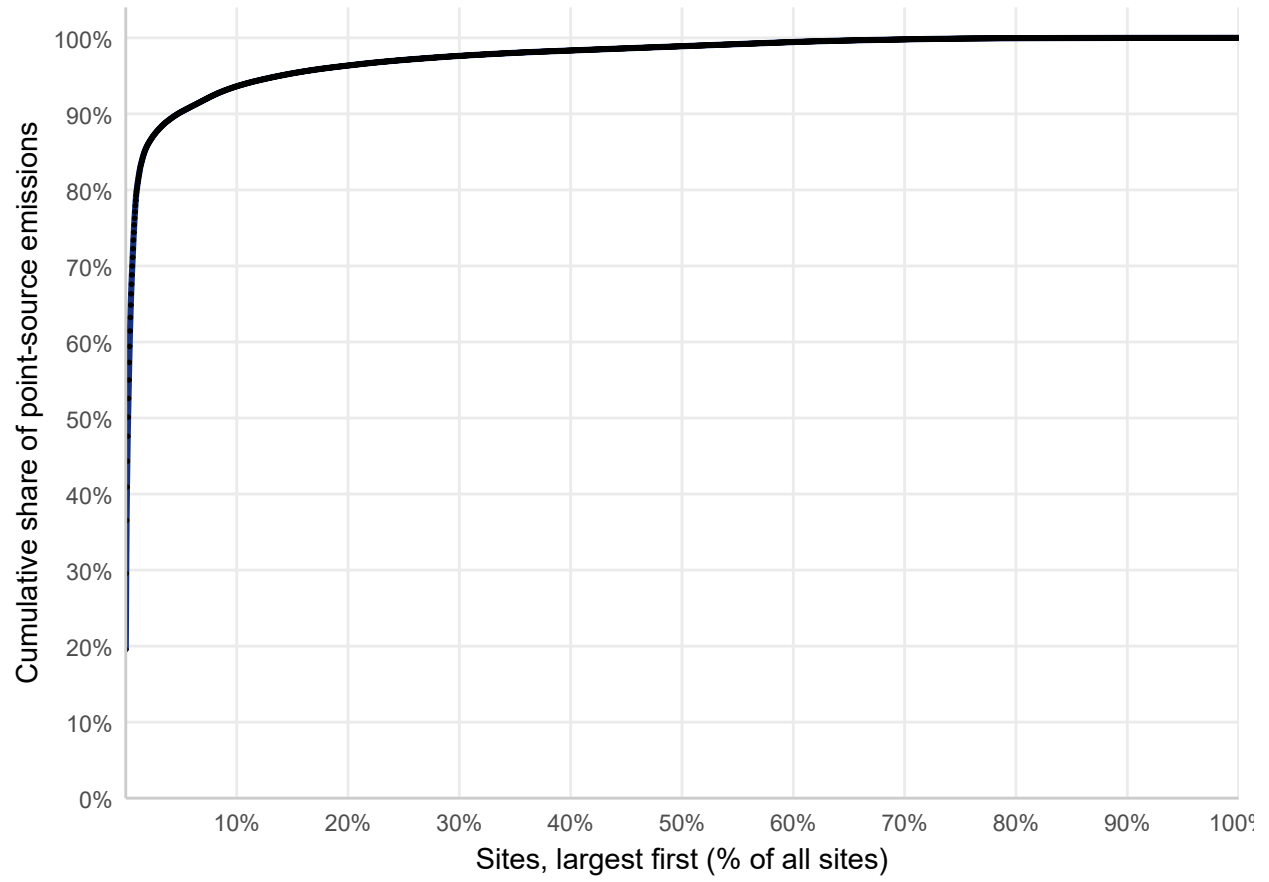
Emissions from Point Sources

Point sources are individual facilities, such as refineries, chemical works, steelworks and incinerators, whose emissions are reported to the NAEI at a specific location. Unlike diffuse sources like road traffic or domestic heating, their emissions are concentrated at one site, so they can have a disproportionate effect on air quality in the surrounding communities.

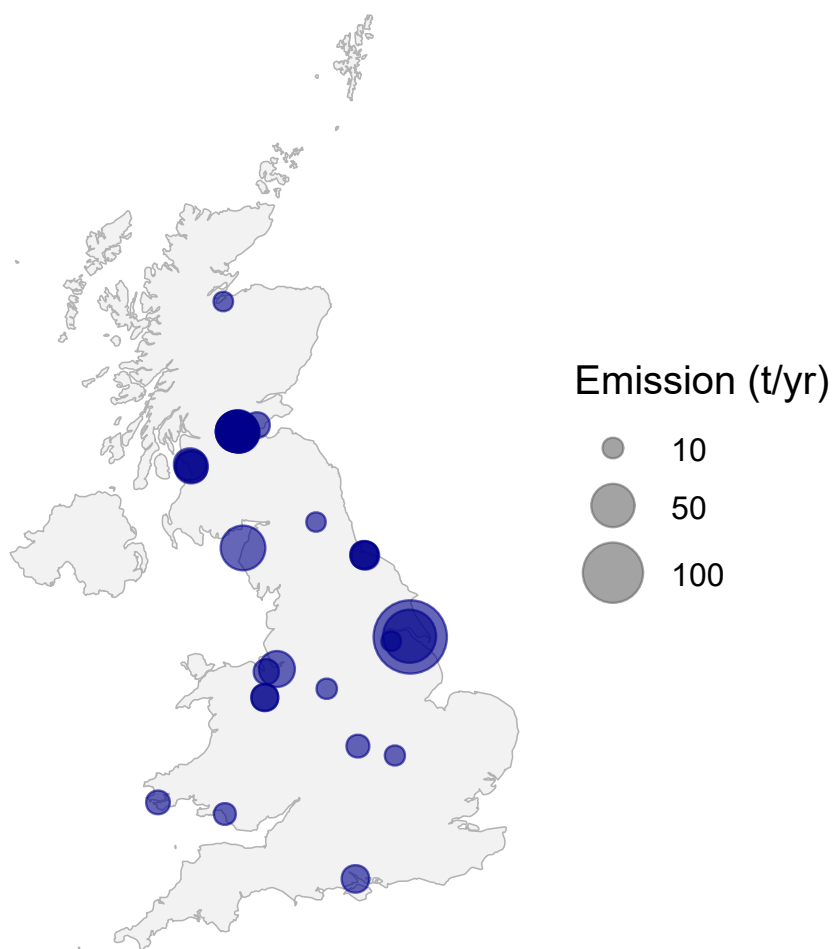
In 2005, point sources accounted for 10.6% of UK benzene emissions (1885 t). As of 2023, point sources accounted for 769 t, representing 9.6% of emissions.



Total point-source emissions are dominated by a small number of sites. Of the 2911 sites reporting benzene emissions in 2023, the ten largest account for 57% of point-source emissions.



Largest point sources



Monitoring

Benzene in ambient air is regulated under the Air Quality Standards Regulations 2010, which set an annual mean limit value of $5 \mu\text{g m}^{-3}$. Defra monitors it through the UK Hydrocarbons Network, which is made up of two complementary parts: a small number of automatic sites giving high time-resolution data, and a larger network of non-automatic samplers giving wider spatial coverage. In 2023, no UK monitoring site exceeded the limit value.

Automatic Hydrocarbon Network

The Automatic Hydrocarbon Network uses automated gas chromatography to measure 29 volatile organic compounds, including benzene and 1,3-butadiene, every hour. It currently has 5 active sites. The hourly data are valuable for understanding daily cycles and short pollution episodes and for identifying sources. However, with so few sites, the network cannot say much about how concentrations vary across the country.

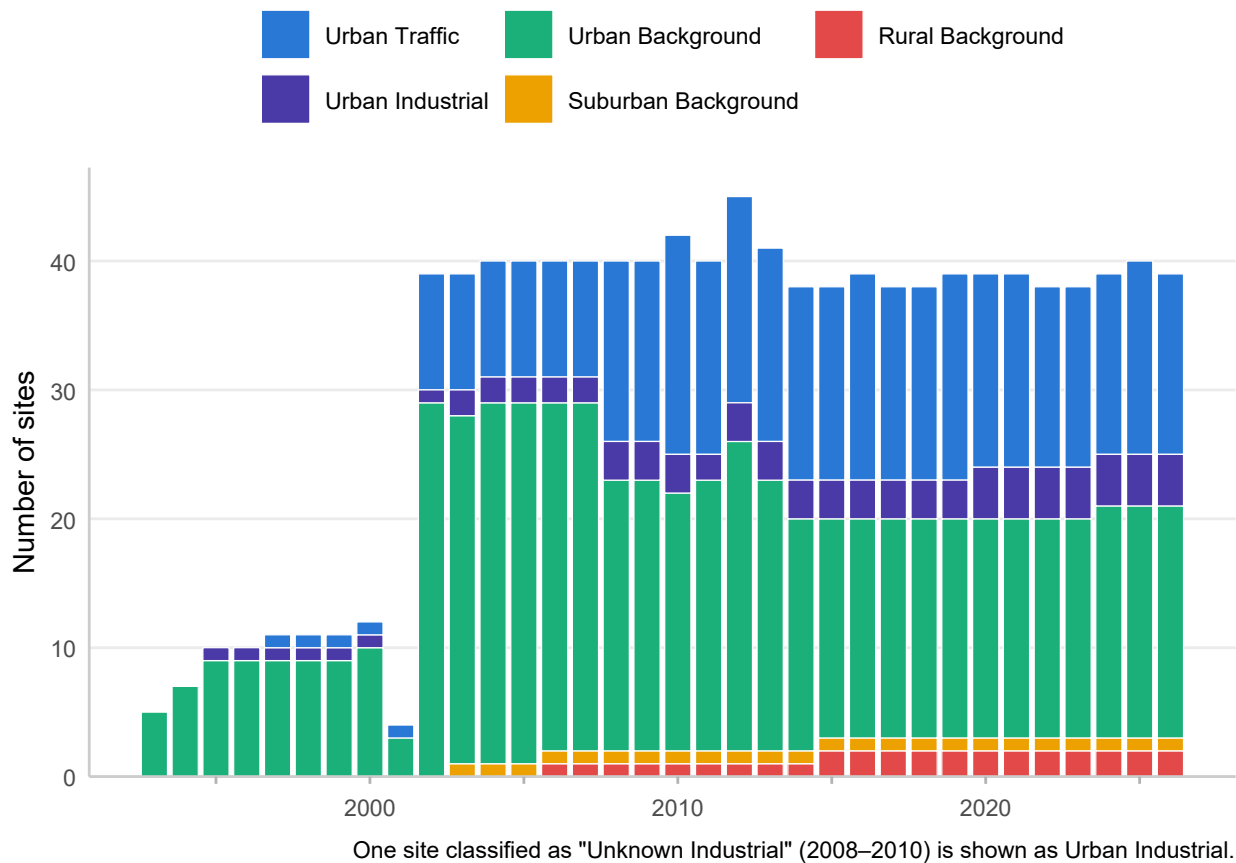
Non-Automatic Hydrocarbon Network

The Non-Automatic Hydrocarbon Network gives much wider coverage, with 34 active sites. Air is drawn through dual sorbent tubes by a controlled-flow pump. The tubes are changed every two weeks and analysed in the laboratory by thermal desorption and gas chromatography–mass spectrometry. This gives fortnightly averages, not hourly values, but the extra sites cover a far wider range of locations.

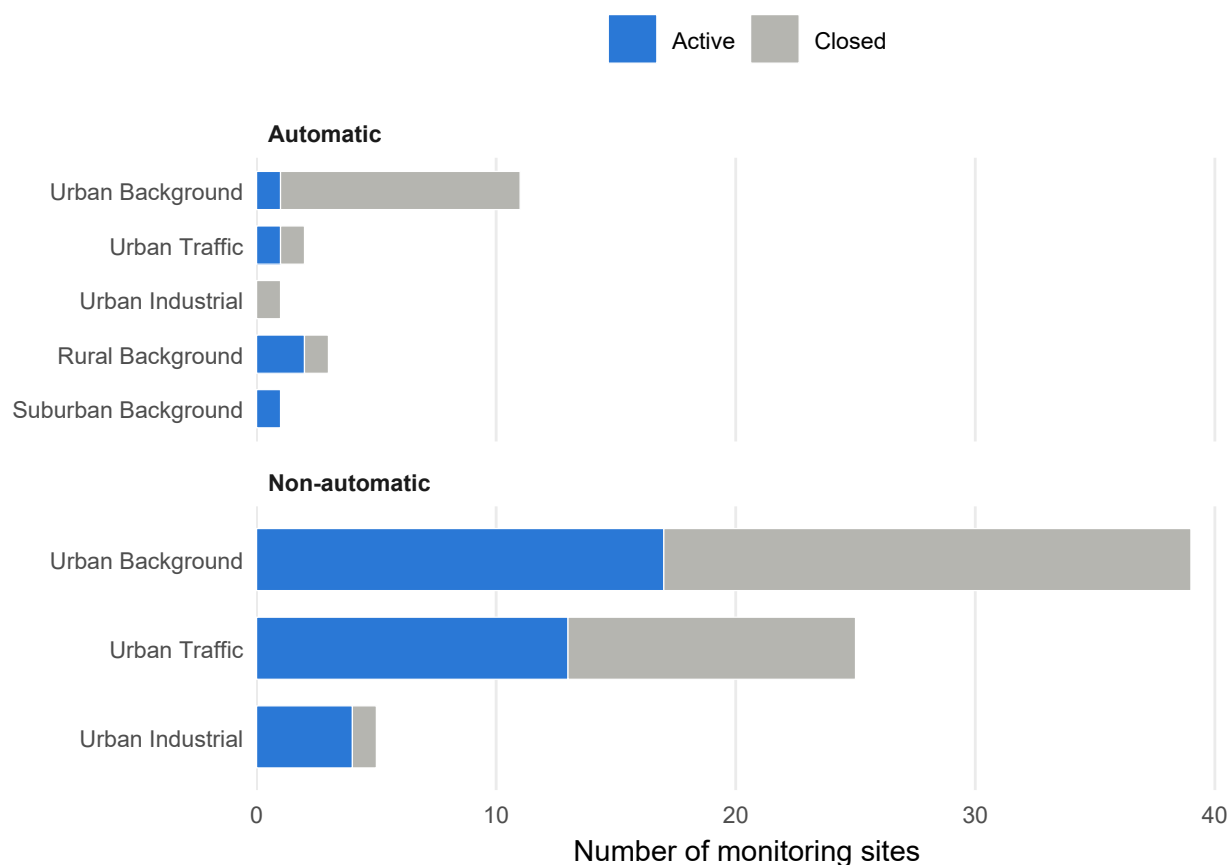
Network coverage over time

87 sites have measured benzene in the two networks at some point, and 39 are active today. Coverage peaked in 2012, with 45 sites running.

Sites per year



Active and closed sites

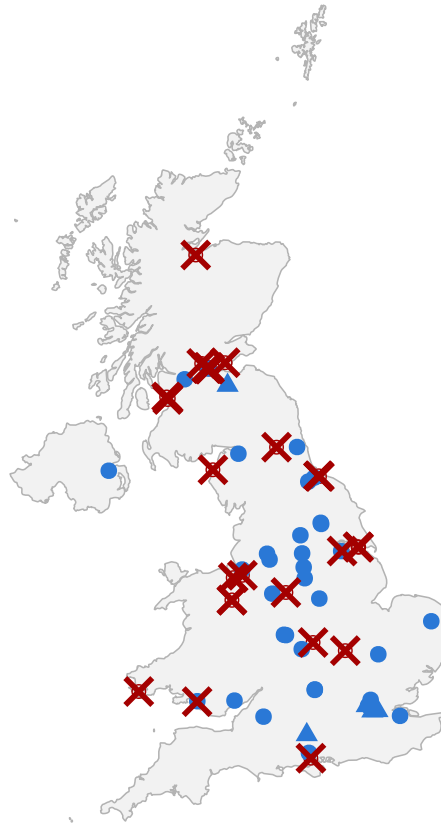


Location of Emissions relative to Monitoring

The monitoring networks were largely designed when road transport dominated benzene emissions, so most sites are in urban areas. As the emission profile has shifted towards domestic combustion and industrial point sources, it is worth asking whether the networks still sample the places where people are most exposed.

Of the 10 largest point sources in 2023, 6 have no active benzene monitor within 10 km. The median distance from one of these sites to its nearest active monitor is 26 km.

Active monitor ▲ Automatic ● Non-automatic



Red crosses: largest point sources, with 10 km radius.

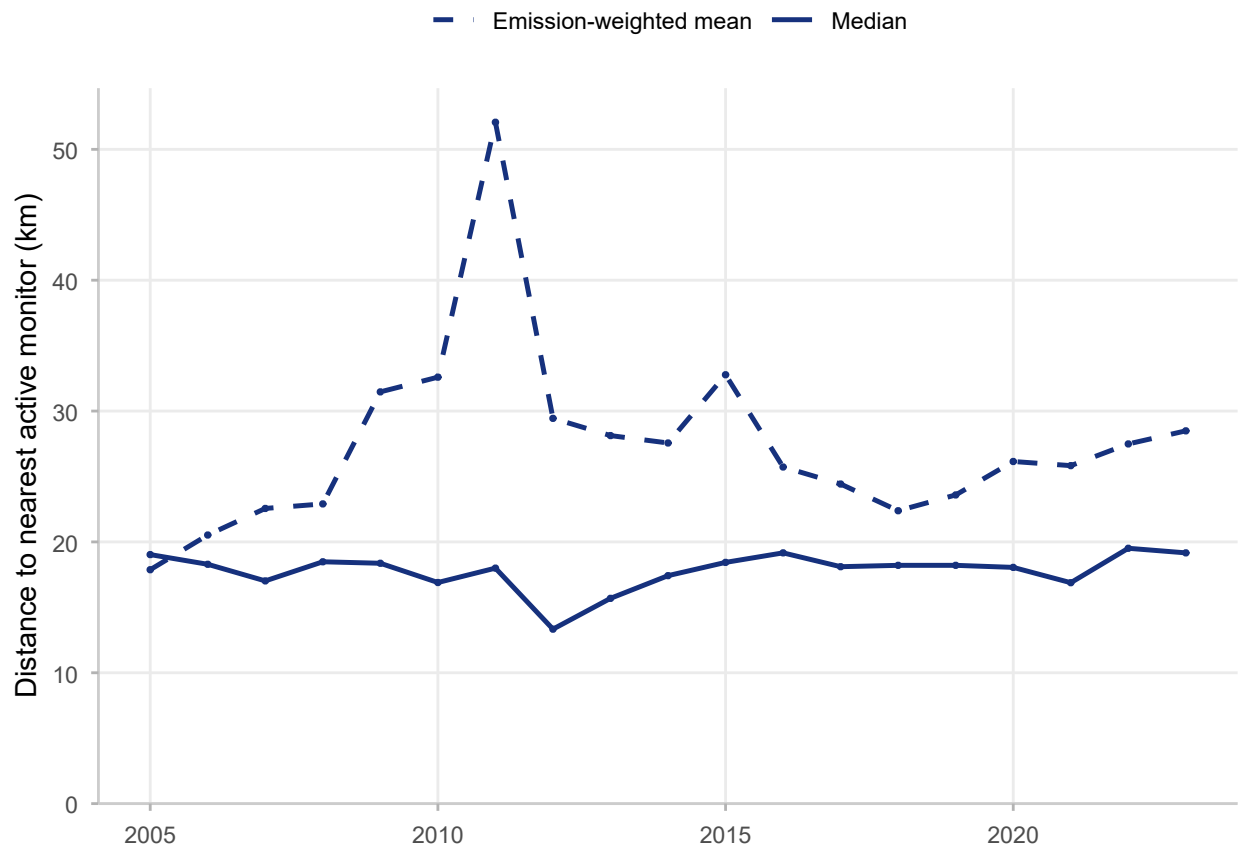
Nearest monitor to each major source

Point source	Emission (t)	Nearest monitor	Monitor type	Distance (km)	Monitors within 10 km
South Killingholme Refinery	150.3	Scunthorpe Town (Non-automatic)	Urban Industrial	26.5	0
Lindsey Oil Refinery	77.0	Scunthorpe Town (Non-automatic)	Urban Industrial	25.8	0
Workington Mill	53.7	Carlisle Morton A595 (Non-automatic)	Urban Traffic	44.6	0
Grangemouth	43.8	Grangemouth (Non-automatic)	Urban Industrial	0.5	1
Stanlow Manufacturing Complex	33.8	Liverpool Speke (Non-automatic)	Urban Industrial	8.5	1
Irvine	26.5	Glasgow Kerbside (Non-automatic)	Urban Traffic	37.4	0

Point source	Emission (t)	Nearest monitor	Monitor type	Distance (km)	Monitors within 10 km
Wilton	19.3	Middlesbrough (Non-automatic)	Urban Back-ground	6.8	1
Fawley Refinery	18.9	Southampton Centre (Non-automatic)	Urban Back-ground	8.5	1
Chirk	17.9	Liverpool Speke (Non-automatic)	Urban Industrial	47.5	0
Mossmorren	16.2	Grangemouth (Non-automatic)	Urban Industrial	26.1	0

Has monitoring kept up with the sources?

The chart below compares each year's 100 largest point sources with the monitors running in that year. It shows the median distance to the nearest monitor, and the mean weighted by emissions, which gives more influence to the largest sources.

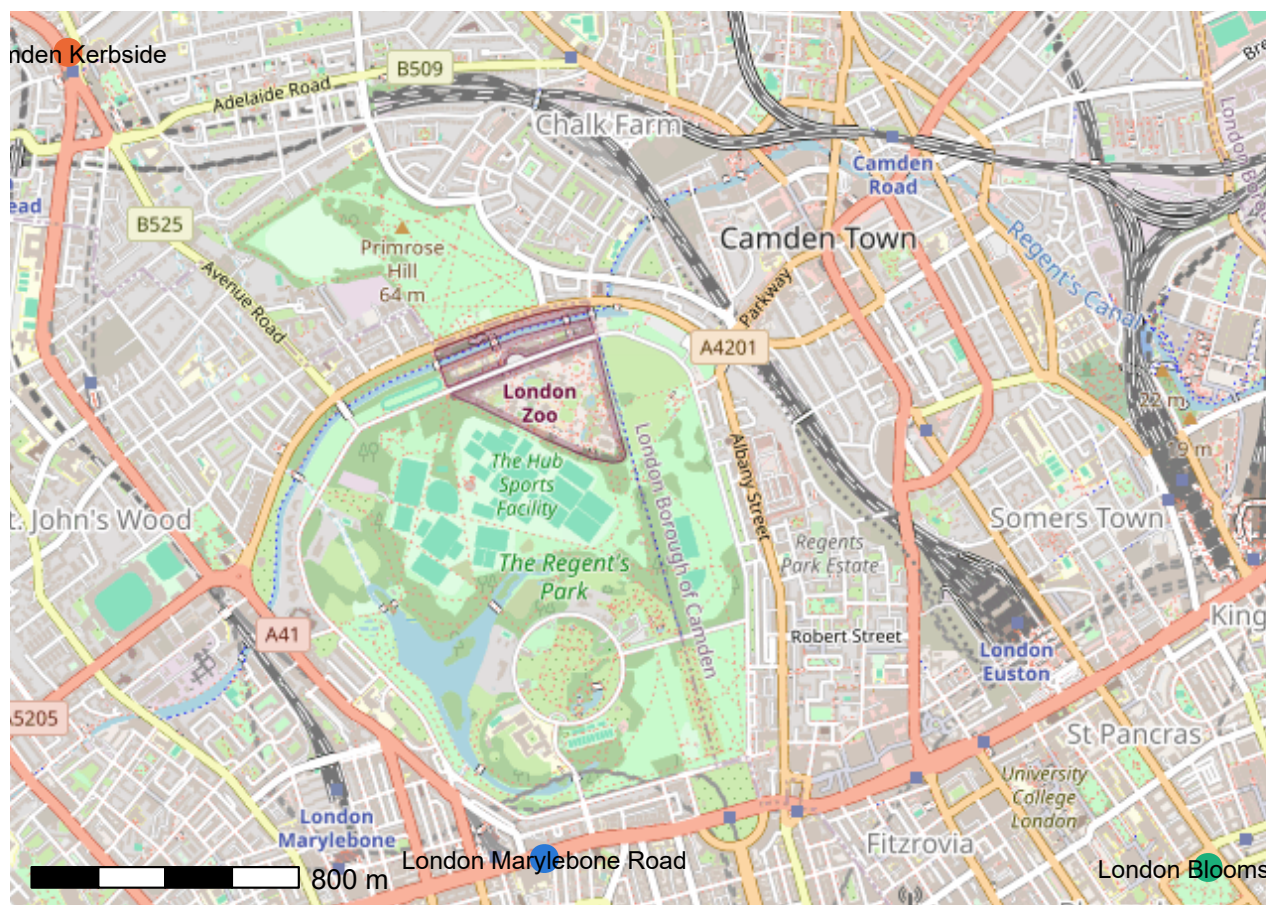


Comparing automatic and non-automatic measurements

No automatic and non-automatic monitors share a location, so the two methods cannot be compared directly. The table below lists the ten closest pairs of automatic and non-automatic sites that operated in the same years.

Automatic site	Type	Non-automatic site	Type	Distance (km)	Years operating together
Edinburgh Med. Sch.	Urban Background	Edinburgh Med. Sch.	Urban Background	0.0	2000–2002 (3 yrs)
Glasgow Kerbside	Urban Traffic	Glasgow Kerbside	Urban Traffic	0.0	2010–2010 (1 yrs)
London Marylebone Road	Urban Traffic	London Bloomsbury	Urban Background	2.0	2002–2026 (25 yrs)
London Marylebone Road	Urban Traffic	Camden Kerbside	Urban Traffic	2.8	2008–2020 (13 yrs)
London Marylebone Road	Urban Traffic	London Farringdon Street	Urban Traffic	3.6	2025–2026 (2 yrs)
London Marylebone Road	Urban Traffic	Tower Hamlets Roadside	Urban Traffic	7.8	2012–2025 (14 yrs)
London Honor Oak Park	Urban Background	Tower Hamlets Roadside	Urban Traffic	8.1	2025–2025 (1 yrs)
London Honor Oak Park	Urban Background	London Farringdon Street	Urban Traffic	8.6	2025–2026 (2 yrs)
London Honor Oak Park	Urban Background	London Bloomsbury	Urban Background	10.1	2025–2026 (2 yrs)
London Marylebone Road	Urban Traffic	Haringey Roadside	Urban Traffic	10.4	2002–2026 (25 yrs)

London Marylebone Road (automatic) was compared with two nearby non-automatic sites: Camden Kerbside, another roadside site, and London Bloomsbury, an urban background site.



The chart below shows fortnightly benzene concentrations at the three sites. The automatic values are hourly measurements averaged over the same two-week periods as the non-automatic samples.

